A Copula-Based Method for Change Detection With Multisensor Optical Remote Sensing Images

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***Abstract*— This article considers the problem of change detection (CD) with multisensor optical remote sensing (RS) images. Copulas are adopted to characterize the dependence structure between the image pair. For this problem, a conditional copula-based CD technique has been proposed in the literature; however, in this technique, it is difficult to select the best copula function in an analytical framework. Resulting copula misspecification may lead to performance degradation. To deal with this problem, we model the CD problem as a binary hypothesis testing problem and propose a new superpixel-level copula-based statistical method (SCOPS) for CD, where an explicit strategy for copula selection is provided for the proposed method. The effectiveness of the copula selection strategy is verified on CD tasks with simulated multisensor optical RS images. Experiments on real RS datasets demonstrate the superiority of SCOPS over state-of-the-art methods.**

***Index Terms*— Change detection (CD), copula theory, remote sensing (RS).**

I. Introduction

CHANGE detection (CD) is an important task in remote sensing (RS), which aims to identify the changes in an area being monitored using RS images acquired at different times. There are plenty of application scenarios of CD, such as disaster assessment, deforestation, urban expansion, and land management [1], [60], [61]. In the last few years, CD has been widely investigated with multitemporal RS images acquired by the same sensor [2], [3], [4], [5], [6], [7], [8], [9], [10].

In practical CD tasks, sometimes, the two images are acquired by different optical sensors. For example, when it is required to reduce the response time to sudden events, such as natural disasters, rapid CD should be carried out by exploiting all available optical sensors. New challenges arise when coping with RS images acquired by different optical sensors, especially considering the significant radiometric differences between the image pair and very dissimilar statistical models corresponding to the two images. In such cases, simple algorithmic operations are no longer able to yield the difference image as done in the CD methods with RS images acquired by the same sensor [11], [28], [29], [30].

Recently, some methods have been proposed, which can be applied to CD problems with multisensor optical RS images[1] [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22]. The existing methods can be classified into three categories: nonparametric methods [12], [13], [14], [15], deep learning methods [16], [17], [18], [19] and parametric methods [20], [21], [22]. In the first category, CD methods proposed in [12] and [13] extract features from the multisensor images and then compute the similarity measures to highlight the changes. In [14], CD is performed by modeling nonlinear dependencies between the images based on local information. Other techniques in this category include transforming the two images to the same domain or to the same feature space to make them comparable using the CD techniques developed for RS images acquired by the same sensor [15]. In the second category, in [17], [18], and [19], image translation techniques are adopted to translate a multisensor image pair into a common space with tools, such as convolutional neural networks (CNNs), conditional generative adversarial networks (cGANs), and cycle-consistent adversarial networks (CycleGANs) to make them comparable to find the changes. Although the nonparametric methods and the deep learning methods attain satisfactory results, their interpretability needs to be improved to enhance situational awareness. Besides, it is time-consuming to run deep learning methods in practical scenarios with limited computational resources. In the third category, statistical tools are employed to model the joint distribution or the statistical dependence between multisensor RS images. These tools include copulas [20], manifold learning [21], and Markov random fields [22]. Compared with nonparametric methods and deep learning methods, parametric methods demonstrate good interpretability and provide analytical guidance for their implementations. Hence, it is appealing to apply parametric methods when possible.

Among the parametric methods, the merit of the copula-based method, i.e., conditional copula-based CD technique (C3D) proposed in [20], lies in being able to model any type of joint distributions of multisensor RS images by decoupling the

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[1]It is worth noting that, in some literature in [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22], RS images acquired by different optical sensors are referred to as heterogenous RS images or multimodal RS images.

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marginal distributions and the dependence structure between the image pair in the unchanged area. In C3D, the local statistics of one image are estimated based on copula theory under the acquisition conditions of the other image. After that, a similarity measure is calculated using Kullback–Leibler divergence to find the changes; however, for C3D, it is difficult to select the best copula function analytically, and copula selection can only be implemented by visual inspection of K-plots. Without any guidance based on theoretical analysis, copula misspecification may occur, which may result in degradation of the CD performance.

Copula theory was originally proposed for modeling the joint distribution of random variables in multivariate analysis [24] and it has been applied in different RS applications such as image classification [51], [52], [53], [54], image fusion [55], and CD [20]. To capture different dependence structures, various copula functions have been constructed [23], [24], [25]. Some frequently used copula functions include Gaussian copula, Student-t copula, Clayton copula, and Gumbel copula [24]. In specific applications, it is necessary to select an appropriate copula function and to estimate its parameters, as done in [26] and [27]. In [26], several methods of copula selection are given for financial applications. In [27], a parametric framework is proposed to deal with the binary hypothesis testing problem in a multisensor system where the observations from different sensors exhibit statistical dependence conditioned on either hypothesis. The optimal copula function is successfully identified in [27], with which the best performance of binary classification could be achieved; however, to the best of our knowledge, the analytical framework for copula selection and binary classification (as in [27]) has not yet been applied in the context of CD with RS images, where copula selection is vital to guarantee a satisfactory CD performance.

For the problem of CD with multisensor optical RS images based on copula theory, to overcome the drawbacks of C3D, this article attempts to propose a new method that selects copulas analytically. Here, we first model the CD problem as a binary hypothesis testing problem and then propose a new superpixel-level copula-based statistical method (SCOPS) for CD. We analyze the CD performance based on the analytical framework in [27] using the features extracted from the superpixels in SCOPS and the test statistics developed for SCOPS. After that, a strategy is provided to select the copula function that attains the best CD performance. To verify the effectiveness of the copula selection strategy in SCOPS, experiments are conducted on simulated RS images, and the performances attained by different copulas are compared to show that the selected copula function achieves an enhanced CD performance. Finally, we run experiments on real RS datasets, which demonstrates the superiority of SCOPS over the state-of-the-art methods.

The main contributions of this article are listed as follows.

1) We model the CD problem with multisensor optical RS images as a binary hypothesis testing problem and propose a new method for CD named SCOPS.
2) In SCOPS that is based on the analytical framework in [27], we derive the relationship between the CD performance and the copula functions based on the features extracted from the superpixels and the test statistic developed for SCOPS, and we provide a strategy for choosing the best copula function from the set of copulas in order to attain the optimal CD performance.
3) We run experiments on a simulated dataset and real RS datasets for CD tasks using SCOPS. Experimental results verify the effectiveness of the copula selection strategy adopted in SCOPS and demonstrate the superiority of the proposed method over the state-of-the-art methods.

We organize the rest of this article as follows. In Section II, we formulate the CD problem with multisensor optical RS images and introduce some related work. In Section III, we construct a statistical model for the considered problem and propose SCOPS, where the strategy for copula selection is specified to attain optimal CD performance. In Section IV, experimental results are shown to verify the effectiveness of the copula selection strategy and to demonstrate the superiority of the proposed method over the state-of-the-art methods. Finally, we present our conclusions in Section V.

II. Problem Formulation and Related Work

A. Problem Formulation

The multisensor optical RS images before and after a certain event are denoted by $\mathbf{X} \sim \mathbb{R}^{M\times N\times C_X}$ and $\mathbf{Y} \sim \mathbb{R}^{M\times N\times C_Y}$, where M and N are the height and the width of the images. The channel numbers of $\mathbf{X}$ and $\mathbf{Y}$ are denoted by C_X and C_Y, which indicates that there could be a single channel or multiple channels in an image, such as different bands in the multispectral images. The two images are acquired by two different optical sensors, respectively, and they are registered before further processing. The pixel intensities in $\mathbf{X}$ and $\mathbf{Y}$ are represented by $x(m,n,c)$ and $y(m,n,c)$, respectively, where $m = 1,\ldots,M$, $n = 1,\ldots,N$, $c = 1,\ldots,C_X$, or $c = 1,\ldots,C_Y$. Let us denote the sets containing all the pixel indices (m,n) in the changed area and in the unchanged area as $\mathcal{C}$ and $\bar{\mathcal{C}}$, respectively, both of which are unknown to us beforehand. The goal of CD is to identify $\mathcal{C}$ and $\bar{\mathcal{C}}$ based on the given images $\mathbf{X}$ and $\mathbf{Y}$ [12], [13], [14], [15], [16], [17], [18], [19], [20], [21], [22].

Similar to the approaches in [15] and [20], a pair of training patches are manually selected from the image pair and they correspond to the same geographic area, which is not affected by the observed event. We represent the training patches from $\mathbf{X}$ and $\mathbf{Y}$ as $\mathbf{X}_{\text{Train}} \sim \mathbb{R}^{M_1\times N_1\times C_X}$ and $\mathbf{Y}_{\text{Train}} \sim \mathbb{R}^{M_1\times N_1\times C_Y}$, respectively, where M_1 and N_1 are the height and width of the training patches. Selecting the training patches requires only a little bit of geographical information regarding the monitored area, and the content of the training patches does not need to be known. For example, if the monitored event is flooding, the changes would occur alongside the river, and the training patches can be acquired from the land areas at a distance from the river. From this perspective, the acquisition of the training patches from the unchanged area requires only very limited prior knowledge [15].

B. Related Work

If $\mathbf{X}$ and $\mathbf{Y}$ are RS images acquired by the same sensor and they lie in a common domain, simple algorithmic operations

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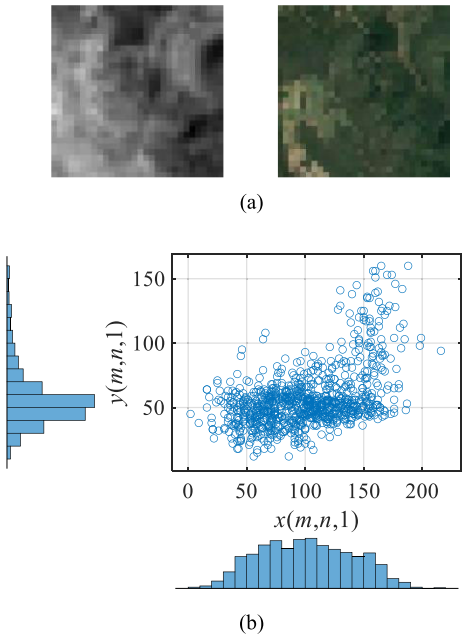


Fig. 1. Illustration of disparate marginal distributions and the statistical dependence structure between the multisensor optical images. (a) Two images obtained by Landsat-5 and Google Earth at two different times at the same location, where change does not occur. (b) Scatter plot of the pixel intensities of the images in (a), where the histograms of the marginal distributions are also shown alongside the axes. We only show the pixel intensities of the first channel ($c = 1$), which is sufficient for illustration.

on them yield the difference image, where the changes can be highlighted [28], [29], [30]. Different from this, although multisensor RS images are taken at the same location and are registered, there are significant radiometric differences and nonlinear variations between them in the unchanged area because the intensities in the two images depend on different kinds of physical properties of the objects. For illustration, Fig. 1(a) provides two images of the same terrain, i.e., $\mathbf{X}$ and $\mathbf{Y}$, taken by Landsat-5 and Google Earth at the same location at different times. Fig. 1(b) shows the scatter plot of $x(m, n, c)$ and $y(m, n, c)$, where $c = 1$. It can be seen that the marginal distributions of $x(m, n, c)$ and $y(m, n, c)$ are disparate, and an intricate dependence structure exists between $x(m, n, c)$ and $y(m, n, c)$.

In the context of CD with multitemporal multisensor optical RS images, to model the dependence structure between the image pair in the presence of dissimilar marginal distributions, copulas are leveraged, and C3D is proposed in [20]; however, in C3D, it is difficult to select the best copula function analytically and any copula misspecification that is present may lead to performance degradation. The proposed method aims to overcome the drawbacks of C3D and improve its CD performance. It is worth noting that substantial differences exist between the proposed method and C3D, which means the former is not a simple extension of the latter. The differences are listed as follows.

1) Contrary to C3D, SCOPS employs a binary hypothesis testing problem to model the CD problem and develops a copula-based test statistic to solve this problem.
2) To avoid copula misspecification, we provide a novel strategy to select an optimal copula function in SCOPS.
3) To maintain the structure of the objects when highlighting the changes and to reduce the computation complexity [12], [31], [32], [33], superpixels are employed as the basic processing units in SCOPS in contrast to [20], which is a pixel-wise method.

III. Statistical Modeling and the Proposed Method

In this section, we first introduce the preprocessing step, i.e., superpixel segmentation and feature extraction. Then, we model the CD problem as a binary hypothesis testing problem and propose SCOPS. The flowchart of SCOPS is presented in Fig. 2.

A. Preprocessing: Superpixel Segmentation and Feature Extraction

Superpixels group the pixels into regions with homogeneous interiors, which contributes to the efficient extraction of features and reduces the negative impact of salt-and-pepper noise in the RS images [32]. Because of its benefits, superpixel segmentation has been widely adopted as the preprocessing step in various RS applications such as image mapping [34], target detection [35], [36], and image classification [37].

In this article, instead of analyzing in a pixel-based manner, SCOPS adopts superpixels as the basic units to detect possible changes. In this way, the different images retain the edges and structures of the objects, which helps to reduce the false negatives and false positives in the changed and unchanged areas. The preprocessing step of SCOPS follows the superpixel co-segmentation[2] procedure in [12]. First, simple linear iterative clustering (SLIC) is implemented on $\mathbf{X}$ and $\mathbf{Y}$, respectively. After that, the superpixel segmentation results of $\mathbf{X}$ and $\mathbf{Y}$ are merged into a cosegmentation map, where the interior of each superpixel is homogeneous in both the preevent image and the postevent image. The superpixel cosegmentation map is represented by $\mathbf{S} \in \mathbb{N}^{M\times N}$, and the elements in $\mathbf{S}$ are denoted by $s(m, n)$, $m = 1, \ldots, M$, $n = 1, \ldots, N$. In $\mathbf{S}$, $s(m, n) = i$ implies that pixel (m, n) belongs to the i-th superpixel and N_s is the total number of superpixels, $i = 1, 2, \ldots, N_s$. Likewise, $\mathbf{X}_{\text{Train}}$ and $\mathbf{Y}_{\text{Train}}$ are cosegmented into N_T superpixels and the cosegmentation map is represented by $\mathbf{S}_1 \in \mathbb{N}^{M_1\times N_1}$.

Based on the superpixel cosegmentation map $\mathbf{S}$, features are extracted from the two images. To be more specific, the mean of the pixel intensities in each channel is calculated to construct the feature vector for each superpixel [12]. In this article, the feature vector of the i-th superpixel in the preevent image $\mathbf{X}$ can be expressed as follows:

$$\mathbf{h}_X(i) = [h_X(i, 1), \ldots, h_X(i, C_X)], \quad i = 1, 2, \ldots, N_s \tag{1}$$

where the channel number of $\mathbf{X}$ is denoted by C_X. In (1)

$$h_X(i, c_1) = \frac{\sum_{m,n} I[s(m, n), i]x(m, n, c_1)}{\sum_{m,n} I[s(m, n), i]} \quad c_1 = 1, \ldots, C_X, \quad i = 1, \ldots, N_s \tag{2}$$

where $I(a, b) = 1$ for $a = b$ and $I(a, b) = 0$ otherwise.

[2]The code is kindly available at https://github.com/yulisun/IRG-McS [12].

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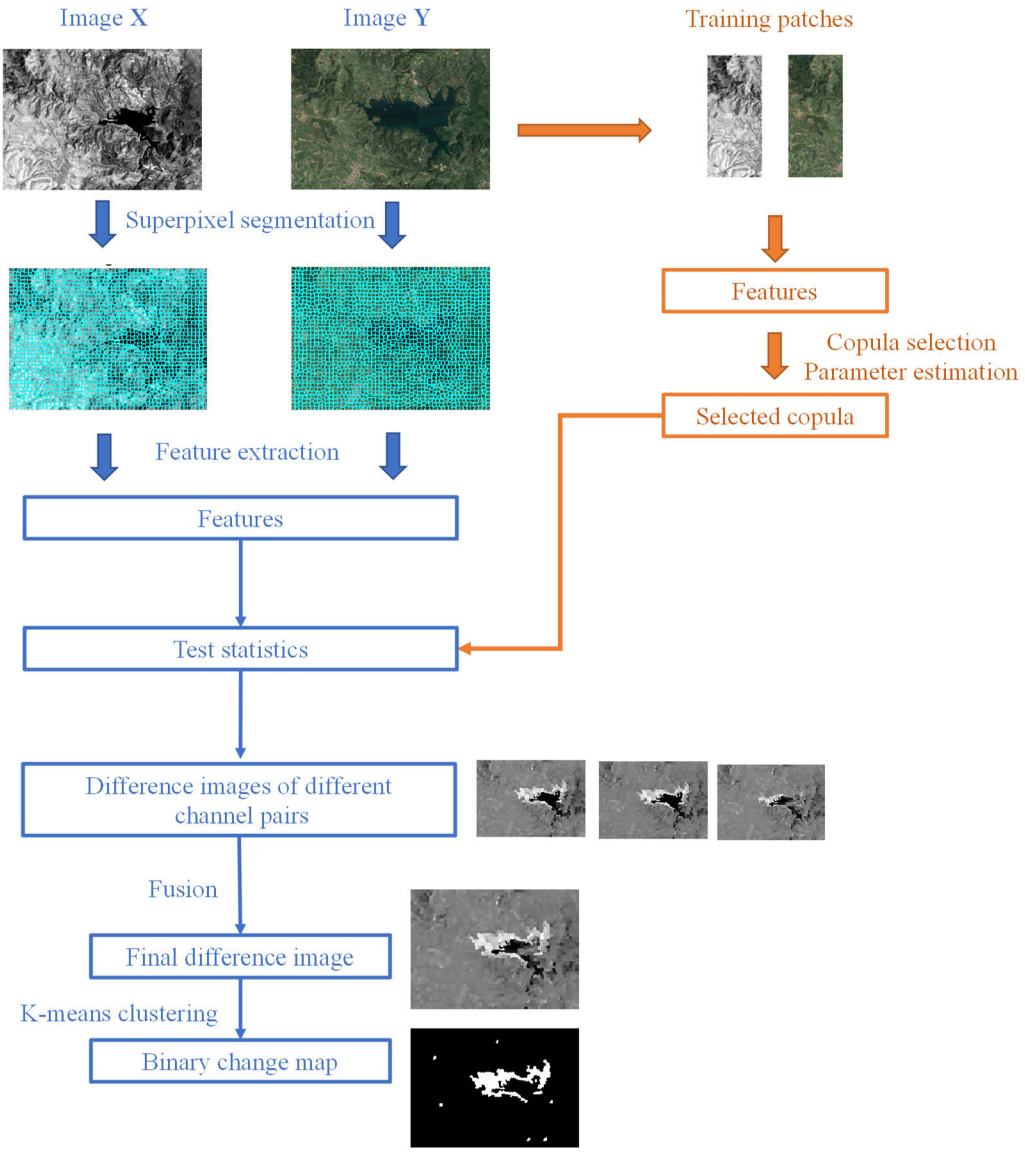


Fig. 2. Flowchart of SCOPS.

Likewise, the feature vector of the i-th superpixel in the postevent image $\mathbf{Y}$ can be expressed as follows:

$$\mathbf{h}_Y(i) = [h_Y(i,1), \ldots, h_Y(i,C_Y)], \quad i = 1, 2, \ldots, N_s \tag{3}$$

where

$$h_Y(i,c_2) = \frac{\sum_{m,n} I[s(m,n),i]\, y(m,n,c_2)}{\sum_{m,n} I[s(m,n),i]} \quad c_2 = 1, \ldots, C_Y, \quad i = 1, \ldots, N_s \tag{4}$$

and the channel number of $\mathbf{Y}$ is denoted by C_Y.

Following similar steps, based on the superpixel cosegmentation map $\mathbf{S}_1 \in \mathbb{N}^{M_1 \times N_1}$ of the training patches, two sets of feature vectors $\{\widehat{\mathbf{h}}_X(i), i = 1, \ldots, N_T\}$ and $\{\widehat{\mathbf{h}}_Y(i), i = 1, \ldots, N_T\}$ can be obtained, where

$$\begin{aligned} \widehat{\mathbf{h}}_X(i) &= \left[\widehat{h}_X(i,1), \ldots, \widehat{h}_X(i,C_X)\right] \\ \widehat{\mathbf{h}}_Y(i) &= \left[\widehat{h}_Y(i,1), \ldots, \widehat{h}_Y(i,C_Y)\right]. \end{aligned} \tag{5}$$

B. Statistical Modeling

For the i-th superpixel under test, with the features extracted from the original images, the CD problem can be modeled as the following binary hypothesis testing problem:

$$\begin{aligned} &\mathrm{H}_0 : h_X(i,c_1), h_Y(i,c_2) \sim f_{XY,c_1,c_2}[h_X(i,c_1), h_Y(i,c_2)] \\ &\mathrm{H}_1 : h_X(i,c_1), h_Y(i,c_2) \sim f^1_{X,c_1}[h_X(i,c_1)] f^1_{Y,c_2}[h_Y(i,c_2)] \end{aligned} \tag{6}$$

with channel c_1 in the preevent image and channel c_2 in the postevent image, $c_1 = 1, \ldots, C_X$, $c_2 = 1, \ldots, C_Y$, $i = 1, \ldots, N_S$. Note that the preevent image and the postevent image are acquired by different optical sensors, which means the channels in the two images have different physical meanings. In (6), the hypothesis H_0 indicates that no change occurs in the i-th superpixel, where statistical dependence exists between $h_X(i,c_1)$ and $h_Y(i,c_2)$. Under H_0, the joint probability density function (pdf) of $h_X(i,c_1)$ and $h_Y(i,c_2)$ is denoted by $f_{XY,c_1,c_2}[h_X(i,c_1), h_Y(i,c_2)]$. The hypothesis H_1

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indicates that the objects contained in the i-th superpixel are different in the two images. In other words, the i-th superpixel is in the changed area. Under H_1, $h_X(i, c_1)$ and $h_Y(i, c_2)$ are statistically independent, and the joint pdf of $h_X(i, c_1)$ and $h_Y(i, c_2)$ is the product of their marginal pdfs, i.e., $f^1_{X,c_1}[h_X(i, c_1)]$ and $f^1_{Y,c_2}[h_Y(i, c_2)]$. Note that the statistical independence between $h_X(i, c_1)$ and $h_Y(i, c_2)$ conditioned on H_1 exists in the cases where the classes of the objects in the i-th superpixel in the changed area of the postevent image do not depend on those of the preevent image. For example, if the monitored event is flooding, in the postevent image, the changed area contains water regardless of the classes of the objects in the preevent image. In this case, the changes can be identified by distinguishing between the dependence structure and the independence structure of the features extracted from the image pair at the same location.

C. Proposed Method: SCOPS

Next, the implementation of SCOPS is introduced, which is composed of three steps: 1) test statistic construction, 2) copula selection and parameter estimation, and 3) binary change map (BCM) generation.

1) Test Statistic Construction: First, we show how to leverage copula theory to capture the joint distribution of the features extracted from the superpixels in the unchanged area of the images and how to construct the test statistic for CD. According to Sklar's Theorem [24], for channel c_1 in the preevent image $\mathbf{X}$ and channel c_2 in the postevent image $\mathbf{Y}$, by denoting the joint cumulative distribution function (cdf) of $h_X(i, c_1)$ and $h_Y(i, c_2)$ under H_0 as $F_{XY,c_1,c_2}[h_X(i, c_1), h_Y(i, c_2)]$, a unique copula function $F_{\text{Copula},c_1,c_2}$ exists such that

$$F_{XY,c_1,c_2}[h_X(i, c_1), h_Y(i, c_2)] = F_{\text{Copula},c_1,c_2}\left\{F^0_{X,c_1}[h_X(i, c_1)], F^0_{Y,c_2}[h_Y(i, c_2)]\right\} \quad \forall i \tag{7}$$

where F^0_{X,c_1} and F^0_{Y,c_2} are the marginal cdfs of $h_X(i, c_1)$ and $h_Y(i, c_2)$ under H_0, $c_1 = 1, \ldots, C_X$, $c_2 = 1, \ldots, C_Y$. In (7), $F_{\text{Copula},c_1,c_2}$ represents the copula function, which is also a joint cdf of two uniform random variables. Taking the second-order derivative of (7), the joint pdf under H_0 can be derived as follows [24]:

$$\begin{aligned} &f_{XY,c_1,c_2}[h_X(i, c_1), h_Y(i, c_2)] \\ &= \frac{\partial^2}{\partial h_X(i, c_1)\partial h_Y(i, c_2)} F_{XY,c_1,c_2}[h_X(i, c_1), h_Y(i, c_2)] \\ &= \frac{\partial^2 F_{\text{Copula},c_1,c_2}\left\{F^0_{X,c_1}[h_X(i, c_1)], F^0_{Y,c_2}[h_Y(i, c_2)]\right\}}{\partial h_X(i, c_1)\partial h_Y(i, c_2)} \\ &= f^0_{X,c_1}[h_X(i, c_1)] f^0_{Y,c_2}[h_Y(i, c_2)] \\ &\quad \times f_{\text{Copula},c_1,c_2}\left\{F^0_{X,c_1}[h_X(i, c_1)], F^0_{Y,c_2}[h_Y(i, c_2)]\right\}, \\ &\forall i \ \forall c_1 \ \forall c_2, \end{aligned} \tag{8}$$

where

$$f_{\text{Copula},c_1,c_2}(u_1, u_2) = \frac{\partial^2 F_{\text{Copula},c_1,c_2}(u_1, u_2)}{\partial u_1 \partial u_2} \tag{9}$$

is the copula density function. In (8), f^0_{X,c_1} and f^0_{Y,c_2} are the marginal pdfs of $h_X(i, c_1)$ and $h_Y(i, c_2)$ under H_0, respectively. Based on (8), it can be seen that, by using copulas to model the joint distribution of multisensor optical RS data in the unchanged area, the complementary information and the common information contained in the image pair can be separated. The marginals embody the complementary information, which is disparate in images acquired by various sensors. The copula density function $f_{\text{Copula},c_1,c_2}$ represents the common information and characterizes the dependence structure existing between the image pair in the unchanged area.

Substituting (8) into (6), the CD problem can be rewritten as follows:

$$\begin{aligned} H_0 &: h_X(i, c_1), h_Y(i, c_2) \\ &\sim f^0_{X,c_1}[h_X(i, c_1)] f^0_{Y,c_2}[h_Y(i, c_2)] \\ &\quad \times f_{\text{Copula},c_1,c_2}\left\{F^0_{X,c_1}[h_X(i, c_1)], F^0_{Y,c_2}[h_Y(i, c_2)]\right\} \\ H_1 &: h_X(i, c_1), h_Y(i, c_2) \\ &\sim f^1_{X,c_1}[h_X(i, c_1)] f^1_{Y,c_2}[h_Y(i, c_2)] \quad \forall i \quad \forall c_1 \forall c_2 \end{aligned} \tag{10}$$

where copulas are leveraged to capture the joint distribution of the features extracted from the superpixels in the unchanged area of the images. By modeling the CD problem as the hypothesis testing problem in (10), CD can be performed at the superpixel level via the optimal log-likelihood ratio (LLR) test. The test statistic can be expressed as follows:

$$\begin{aligned} T_{i,c_1,c_2} = &-\log f^0_{X,c_1}[h_X(i, c_1)] - \log f^0_{Y,c_2}[h_Y(i, c_2)] \\ &- \log f_{\text{Copula},c_1,c_2}\left\{F^0_{X,c_1}[h_X(i, c_1)], F^0_{Y,c_2}[h_Y(i, c_2)]\right\} \\ &+ \log f^1_{X,c_1}[h_X(i, c_1)] + \log f^1_{Y,c_2}[h_Y(i, c_2)] \underset{H_0}{\overset{H_1}{\gtrless}} \gamma \end{aligned} \tag{11}$$

where γ is the decision threshold, $\forall i, \forall c_1, \forall c_2$. In (11), the marginal pdfs $\{f^0_{X,c_1}, f^0_{Y,c_2}\}$ can be modeled by Gaussian mixture distribution with Q_{GM} components, whose parameters can be estimated with the training patches.[3] The marginal cdfs $\{F^0_{X,c_1}, F^0_{Y,c_2}\}$ in the unchanged area can be estimated empirically via the canonical maximum likelihood method (CML) [50] as follows:

$$F^0_{X,c_1}(h) \approx \frac{1}{N_T}\sum_{i=1}^{N_T} \chi\left[h - \widehat{h}_X(i, c_1)\right] \quad \forall c_1 \tag{12}$$

and

$$F^0_{Y,c_2}(h) \approx \frac{1}{N_T}\sum_{i=1}^{N_T} \chi\left[h - \widehat{h}_Y(i, c_2)\right] \quad \forall c_2 \tag{13}$$

where $\chi(\cdot)$ is the step function and $\{\widehat{h}_X(i, c_1), \widehat{h}_Y(i, c_2)\}$ are obtained from the training patches, as shown in Section III-A. The copula density function $f_{\text{Copula},c_1,c_2}$ can be selected from the set of copulas [24] given later in the article.[4] The selection strategy will be specified later in this section. It is, however, difficult to estimate the marginal pdfs $\{f^1_{X,c_1}, f^1_{Y,c_2}\}$ due to the absence of training data for the changed area.

[3]The marginal pdfs can be modeled by other statistical distributions as well, but it does not affect the basic idea behind SCOPS.

[4]Based on (9), selection of the copula density function $f_{\text{Copula},c_1,c_2}$ is equivalent to the selection of the copula function $F_{\text{Copula},c_1,c_2}$.

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To derive the test statistic T_{i,c_1,c_2} in this case, inspired by the theory of Bayesian statistical inference [38], we provide four priors on the features extracted from the changed area, which characterize different beliefs regarding possible changes according to relevant evidence and subjective assessments. We will provide some hints on how to select an appropriate prior based on the experimental results in Section IV-B.

The priors are given as follows:

Prior 1: $f^0_{X,c_1} \approx f^1_{X,c_1}$, $f^0_{Y,c_2} \approx f^1_{Y,c_2}$;
Prior 2: $f^1_{X,c_1} =$ constant, $f^1_{Y,c_2} =$ constant;
Prior 3: $f^0_{X,c_1} \approx f^1_{X,c_1}$, $f^1_{Y,c_2} =$ constant; and
Prior 4: $f^1_{X,c_1} =$ constant, $f^0_{Y,c_2} \approx f^1_{Y,c_2}$.

Prior 1 is the most informative prior, which assumes that the marginal pdfs of the features in the changed area and in the unchanged area are the same in both the preevent image and the postevent image. Prior 1 can be applied when we hold the belief that, in both images, the changed area and the unchanged area contain objects of the same type with the same properties. Prior 2 is the most uninformative prior, which adopts flat priors and conveys very vague information [39], [56], [57]. Prior 3 and Prior 4 are weakly informative priors. When applying them, we hold the belief that in one of the images, the changed area and the unchanged area contain objects of the same type [20]. In contrast, in the other image, we only have general and vague information about the changed area and consequently adopt the flat prior [39], [56], [57]. For example, if the event being monitored is flooding, the changed area and the unchanged area in the preevent image are likely to contain the same type of objects, such as trees, houses, and crops. In this case, it is rational to hold the belief that $f^0_{X,c_1} \approx f^1_{X,c_1}$ and to apply Prior 3.

Given the four priors, the test statistic in (11) can be rewritten as follows:

$$T^1_{i,c_1,c_2} = -\log f_{\text{Copula},c_1,c_2}\left\{F^0_{X,c_1}[h_X(i,c_1)], F^0_{Y,c_2}[h_Y(i,c_2)]\right\} \tag{14}$$

$$T^2_{i,c_1,c_2} = -\log f^0_{X,c_1}[h_X(i,c_1)] - \log f^0_{Y,c_2}[h_Y(i,c_2)] - \log f_{\text{Copula},c_1,c_2}\left\{F^0_{X,c_1}[h_X(i,c_1)], F^0_{Y,c_2}[h_Y(i,c_2)]\right\} \tag{15}$$

$$T^3_{i,c_1,c_2} = -\log f^0_{Y,c_2}[h_Y(i,c_2)] - \log f_{\text{Copula},c_1,c_2}\left\{F^0_{X,c_1}[h_X(i,c_1)], F^0_{Y,c_2}[h_Y(i,c_2)]\right\} \tag{16}$$

$$T^4_{i,c_1,c_2} = -\log f^0_{X,c_1}[h_X(i,c_1)] - \log f_{\text{Copula},c_1,c_2}\left\{F^0_{X,c_1}[h_X(i,c_1)], F^0_{Y,c_2}[h_Y(i,c_2)]\right\} \tag{17}$$

respectively. From (14) to (17), under Prior p, for channel c_1 in the preevent image and channel c_2 in the postevent image, the test statistics computed for all the superpixels in the images are given in the following set:

$$\left\{T^p_{i,c_1,c_2}, c_1 = 1,\ldots,C_X, c_2 = 1,\ldots,C_Y, i = 1,\ldots,N_S\right\} \quad p = 1,2,3,4. \tag{18}$$

2) Copula Selection and Parameter Estimation: Next, we capture the CD performance of the test statistic obtained as (18) and provide the strategy for copula selection. In [27], the effect of copula misspecification on the performance of binary hypothesis testing is analyzed. Based on the analytical framework of [27], with the model in (10) and the test statistic in (11), if the copula density function $f_{\text{Copula},c_1,c_2}$ is misspecified as $\hat{f}_{\text{Copula},c_1,c_2}$, the test statistic is given as follows:

$$\hat{T}_{i,c_1,c_2} = -\log f^0_{X,c_1}[h_X(i,c_1)] - \log f^0_{Y,c_2}[h_Y(i,c_2)] - \log \hat{f}_{\text{Copula},c_1,c_2}\left\{F^0_{X,c_1}[h_X(i,c_1)], F^0_{Y,c_2}[h_Y(i,c_2)]\right\} + \log f^1_{X,c_1}[h_X(i,c_1)] + \log f^1_{Y,c_2}[h_Y(i,c_2)] \underset{H_0}{\overset{H_1}{\gtrless}} \gamma. \tag{19}$$

Implementing (19) instead of (11) in the context of CD, the performance loss can be expressed as follows [40]:

$$\Delta = \int \left\{P_U(\gamma) - \hat{P}_U(\gamma)\right\} d\gamma = D\left(f_{XY,c_1,c_2}, \hat{f}_{XY,c_1,c_2}\right) \tag{20}$$

where

$$\hat{f}_{XY,c_1,c_2}(h_1,h_2) = f^0_{X,c_1}(h_1) f^0_{Y,c_2}(h_2) \hat{f}_{\text{Copula},c_1,c_2}\left[F^0_{X,c_1}(h_1), F^0_{Y,c_2}(h_2)\right]. \tag{21}$$

In (20), $P_U(\gamma) = \Pr(T_{i,c_1,c_2} < \gamma | \mathrm{H}_0)$ is the probability of correctly identifying the i-th superpixel as a superpixel in the unchanged area when $f_{\text{Copula},c_1,c_2}$ is perfectly known, $\hat{P}_U(\gamma) = \Pr(\hat{T}_{i,c_1,c_2} < \gamma | \mathrm{H}_0)$ is the probability of correctly identifying the i-th superpixel as a superpixel in the unchanged area when $f_{\text{Copula},c_1,c_2}$ is misspecified as $\hat{f}_{\text{Copula},c_1,c_2}$, and $D(f_{XY,c_1,c_2}, \hat{f}_{XY,c_1,c_2})$ is the Kullback–Leibler divergence between f_{XY,c_1,c_2} and $\hat{f}_{XY,c_1,c_2}$.

Based on (20), to reduce the performance loss so as to optimize the CD performance, $\hat{f}_{XY,c_1,c_2}$ should be selected, which minimizes $D(f_{XY,c_1,c_2}, \hat{f}_{XY,c_1,c_2})$. Equivalently, the selected copula density function satisfies [27]

$$\hat{f}^{\text{opt}}_{\text{Copula},c_1,c_2} = \underset{\hat{f}_{\text{Copula},c_1,c_2}}{\arg\max} \int_{f_{XY,c_1,c_2}} (h_1,h_2) \times \log \hat{f}_{\text{Copula},c_1,c_2}\left[F^0_{X,c_1}(h_1), F^0_{Y,c_2}(h_2)\right] dh_1 dh_2, \quad \forall c_1\ \forall c_2. \tag{22}$$

Next, we show how to solve (22) with the training patches to find the best copula density function. Considering that the training data $\mathbf{X}_{\text{Train}}$ and $\mathbf{Y}_{\text{Train}}$ are randomly selected from the unchanged area of $\mathbf{X}$ and $\mathbf{Y}$, it can be rationally assumed that the statistical distributions of the features extracted from the superpixels in the training patches are very similar to those in the unchanged area of $\mathbf{X}$ and $\mathbf{Y}$. Based on (5), we can approximate (22) as follows [27]:

$$\hat{f}^{\text{opt}}_{\text{Copula},c_1,c_2} = \underset{\hat{f}_{\text{Copula},c_1,c_2}}{\arg\max} \frac{1}{N_T} \sum_{i=1}^{N_T} \log \hat{f}_{\text{Copula},c_1,c_2}\left\{F^0_{X,c_1}\left[\widehat{h}_X(i,c_1)\right], F^0_{Y,c_2}\left[\widehat{h}_Y(i,c_2)\right]\right\}, \tag{23}$$

where F^0_{X,c_1} and F^0_{Y,c_2} are estimated empirically via (12) and (13). In practice, an optimal copula density function

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is selected from the set of copula density functions $\Theta = \{f_1, \ldots, f_Q\}$, each of which is parameterized by $\boldsymbol{\theta}_q$, $q = 1, \ldots, Q$. Among the set Θ, the best copula density function is given as follows:

$$\hat{f}^{\text{opt}}_{\text{Copula},c_1,c_2} = f_{q_{\text{opt}}}\left(u_1, u_2; \boldsymbol{\theta}^{\text{opt}}_{q_{\text{opt}}}\right) \quad \forall c_1 \ \forall c_2 \tag{24}$$

where

$$\begin{aligned} q_{\text{opt}} &= \arg\max_q g_q \\ g_q &= \frac{1}{N_T}\sum_{i=1}^{N_T} \log f_q\left\{F^0_{X,c_1}\left[\widehat{h}_X(i,c_1)\right], F^0_{Y,c_2}\left[\widehat{h}_Y(i,c_2)\right]; \boldsymbol{\theta}^{\text{opt}}_q\right\} \\ \boldsymbol{\theta}^{\text{opt}}_q &= \arg\max_{\boldsymbol{\theta}_q} \frac{1}{N_T}\sum_{i=1}^{N_T} \log f_q\left\{F^0_{X,c_1}\left[\widehat{h}_X(i,c_1)\right], \right. \\ &\qquad \left. F^0_{Y,c_2}\left[\widehat{h}_Y(i,c_2)\right]; \boldsymbol{\theta}_q\right\} \end{aligned} \tag{25}$$

and $f_q = f_q(u_1, u_2; \boldsymbol{\theta}_q)$ is the q-th copula density function in Θ. Substituting (24) into (11) yields the test statistic (18).

3) BCM Generation: With the CD problem modeled as (10), by thresholding the LLR test statistic at the superpixel level, as shown in (18), a total number of $C_X C_Y$ BCMs are obtained from different channel pairs of the images. Although a final BCM can be obtained by fusing those BCMs directly, it induces severe information loss. In SCOPS, an alternative method is adopted to fuse the information from different channel pairs. First, the test statistics of different channel pairs are fused to generate the final different image $\text{DI}^p = \{\text{DI}^p_i, \forall i\}$, where

$$\text{DI}^p_i = \max\left\{T^p_{i,c_1,c_2}, \forall c_1, \forall c_2\right\} \quad \forall i \ \forall p. \tag{26}$$

In the final different image DI^p, the changes highlighted by any two channels $\{c_1, c_2\}$ of the image pair are maintained, which contributes to the reduction of false negatives of CD.

Based on the different image DI^p, next we show how to generate the BCM. A straightforward way is to directly threshold DI^p to find the changes; however, it is difficult to determine the threshold subject to a trade-off between the false positives and false negatives. To cope with this problem, K-means clustering is implemented as follows to derive the BCM, leveraging the pixel intensities and the final different image simultaneously.

Let us represent the i-th superpixel by a vector

$$\begin{aligned} \mathbf{v}_i = [&h_X(i,1), \ldots, h_X(i,C_X), h_Y(i,1), \ldots, \\ &h_Y(i,C_Y), \alpha \tilde{DI}^p_i] \quad \forall i \end{aligned} \tag{27}$$

where $\tilde{DI}^p_i$ is the normalized different image given as follows:

$$\tilde{DI}^p_i = \text{DI}^p_i - \Lambda\left(\text{DI}^p\right) \quad \forall i \tag{28}$$

and α is the weight of the different image during the process of clustering. In (28), $\Lambda(\cdot)$ is the mean function, whose output is the mean value of all the elements in the input matrix.

Next, $\{\mathbf{v}_i, i = 1, \ldots, N_s\}$ is clustered into two clusters, i.e., $\mathcal{D}_1$ and $\mathcal{D}_2$, via K-means clustering, which includes a cluster of superpixels in the changed area and a cluster of superpixels in the unchanged area. Let us denote the set containing all the indices of the superpixels in the changed area as $\mathcal{D}_C$, which can be determined as follows:

$$\mathcal{D}_C = \arg\max_{\mathcal{D}\subset\{\mathcal{D}_1,\mathcal{D}_2\}} \frac{1}{|\mathcal{D}|}\sum_{i\in\mathcal{D}} \tilde{\text{DI}}^p_i. \tag{29}$$

In (29), it is implied that $\mathcal{D}_C$ is selected as the cluster where the superpixels therein have larger values of $\tilde{\text{DI}}^p_i$. Finally, the BCM $\mathbf{B} \sim \mathbb{N}^{M\times N}$ is generated as follows:

$$\begin{aligned} b(m,n) &= 1, \quad \text{if } s(m,n) = i, \ i \in \mathcal{D}_C \\ b(m,n) &= 0, \quad \text{else} \end{aligned} \tag{30}$$

where $b(m,n)$ is the (m,n)-th element in $\mathbf{B}$.

TABLE I
COPULA FUNCTIONS IN THE GIVEN SET

Copula	$F_{Copula}(u_1,u_2)$
Gaussian	$\phi_\rho\left(\phi^{-1}(u_1), \phi^{-1}(u_2)\right)$, ϕ_ρ is the cdf of bivariate standard normal distribution with correlation ρ, and ϕ is the cdf of univariate standard normal distribution.
Student-t	$t_{\rho,v}\left(t_v^{-1}(u_1), t_v^{-1}(u_2)\right)$, $t_{\rho,v}$ is the cdf of bivariate standard Student-t distribution with correlation ρ and degrees of freedom v, and t_v is the cdf of univariate standard Student-t distribution with degrees of freedom v.
Clayton	$\left(u_1^{-\theta} + u_2^{-\theta} - 1\right)^{-\frac{1}{\theta}}$
Gumbel	$\exp\left[-\left(\left(-\log u_1\right)^\theta + \left(-\log u_2\right)^\theta\right)^{\frac{1}{\theta}}\right]$

IV. EXPERIMENTAL RESULTS

To validate the analysis in this article and demonstrate the superiority of SCOPS, we run experiments on simulated RS images and real RS datasets. In these experiments, a given set of candidate copulas is considered, which includes the Gaussian copula, Student-t copula, Clayton copula, and Gumbel copula.[5] The copula functions in the given set are listed in Table I.

A. Experiments on Simulated RS Images

We conduct experiments on a pair of simulated multisensor optical RS images and compare the performances attained by different copulas in order to verify that a better CD performance can be attained by the copula function chosen by SCOPS compared with other candidate copulas.

[5]It is well worth noting that the proposed method can be applied on any given set of copula functions.

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Considering that the real images are composed of different superpixels with homogeneous interiors, the simulated RS images are generated by defining a superpixel cosegmentation map and assigning random values to the superpixels, which represent features at the superpixel level. To be more specific, as the first step, a superpixel cosegmentation map of size 2336×3504 is randomly generated with $N_s = 5000$ superpixels. Among all the superpixels, 251 superpixels are in the changed area and their indices are contained in set $\mathcal{A}$. The remaining superpixels are in the unchanged area, whose indices are contained in set $\bar{\mathcal{A}}$. After that, two sets of random variables denoting the features at the superpixel level of the two images, i.e., $\{z_1(i), i = 1, \ldots, N_s\}$ and $\{z_2(i), i = 1, \ldots, N_s\}$, are generated based on the statistical model constructed as (6), which reflect the physical properties of the objects in the images. The marginal pdfs are $z_1(i) \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $z_2(i) \sim \mathcal{N}(\mu_2, \sigma_2^2)$, $\forall i$. For $i \in \bar{\mathcal{A}}$, the i-th superpixel is in the unchanged area and the dependence structure between features $z_1(i)$ and $z_2(i)$ is captured by Clayton copula with parameter θ. Coupling the marginal pdfs of $\{z_1(i), z_2(i)\}$ and the Clayton copula with parameter θ yields the joint pdf. For $i \in \mathcal{A}$, the i-th superpixel is in the changed area and the joint pdf of features $\{z_1(i), z_2(i)\}$ is the product of their marginal pdfs. Under the above setting, statistical dependence between $z_1(i)$ and $z_2(i)$ exists if and only if the i-th superpixel is in the unchanged area. Next, pixel intensities in image $\mathbf{X}$ and $\mathbf{Y}$ are generated based on $z_1(i)$ and $z_2(i)$, respectively. For optical RS images, it is rational to assume that the pixel intensity is corrupted by an additive Gaussian noise. To be more specific, if pixel (m, n) belongs to the i-th superpixel, its intensity in image $\mathbf{X}$ is

$$x(m, n, 1) = z_1(i) + z_{1,\text{Noise}}(m, n) \quad \forall m \ \forall n \ \forall i \tag{31}$$

where $z_{1,\text{Noise}}(m, n) \sim \mathcal{N}(0, \sigma_N^2)$ is the additive Gaussian noise independently distributed across different pixels. Likewise, the intensity of pixel (m, n) in the i-th superpixel in image $\mathbf{Y}$ is

$$y(m, n, 1) = z_2(i) + z_{2,\text{Noise}}(m, n) \quad \forall m \ \forall n \ \forall i \tag{32}$$

where $z_{2,\text{Noise}}(m, n) \sim \mathcal{N}(0, \sigma_N^2)$.

Following the above steps, we can obtain the preevent image $\mathbf{X}$ and the postevent image $\mathbf{Y}$ of size $2336 \times 3504 \times 1$. They are cosegmented into $N_s = 5000$ superpixels, which consist of the changed area and the unchanged area. The features extracted from the superpixels in the changed area exhibit a dependence structure characterized by Clayton copula in the image pair, while in the unchanged area, they are independent. From the generation process of the simulated RS images, it can be seen that the marginal pdfs of the features extracted from the superpixels in the changed area and in the unchanged area are the same, which is in accordance with Prior 1.

In the experiments in Section IV-A, we fix $\mu_1 = 1$, $\sigma_1 = 2$, $\mu_2 = 2$, $\sigma_2 = 4$, and $\sigma_N = 0.5$. In $\mathbf{X}$ and $\mathbf{Y}$, we designate $\lambda\%$ of the data as training data, with the remaining serving as test data. Here, $\vartheta\%$ of the training data originate from the changed area, while $(100 - \vartheta)\%$ of the training data come from the unchanged area. By default, we set $\vartheta = 0$, which means all the training data are selected from the unchanged area. This selection process can be achieved by incorporating geographical information, as mentioned in Section II-A. For

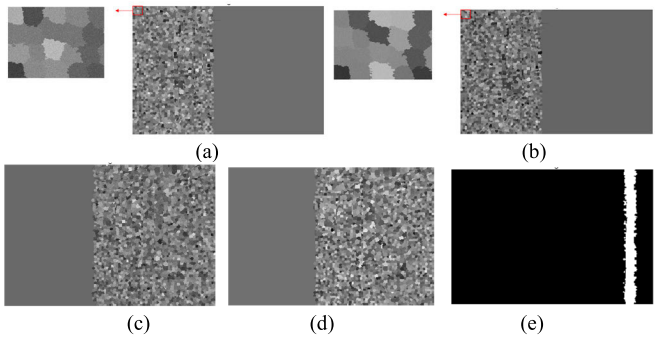


Fig. 3. Pair of simulated multisensor RS images. (a) Training data in the preevent image. (b) Training data in the postevent image. (c) Test data in the preevent image. (d) Test data in the postevent image. (e) Ground truth of the change mask.

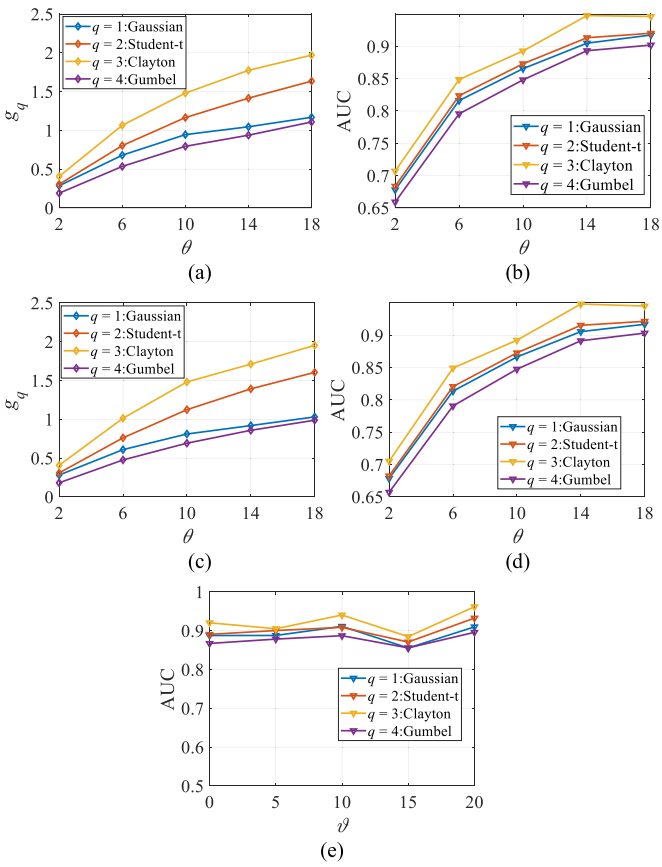


Fig. 4. Comparison among different copula functions. (a) $\{g_q, q = 1, 2, 3, 4\}$ as a function of θ, where $\lambda = 40$ and $\vartheta = 0$. (b) AUC value as a function of θ, where $\lambda = 40$ and $\vartheta = 0$. (c) $\{g_q, q = 1, 2, 3, 4\}$ as a function of θ, where $\lambda = 20$ and $\vartheta = 0$. (d) AUC value as a function of θ, where $\lambda = 20$ and $\vartheta = 0$. (e) AUC value as a function of ϑ, where $\lambda = 20$ and $\theta = 10$.

illustration, when the parameters are set as $\theta = 2$, $\lambda = 40$ and $\vartheta = 0$, the simulated images are shown in Fig. 3, where the training data are presented in Fig. 3(a) and (b), and the test data are presented in Fig. 3(c) and (d). The ground truth of the change mask is shown in Fig. 3(e). Fig. 3 reveals that an area in $\mathbf{X}$ can be brighter or darker in $\mathbf{Y}$ at the same location, illustrating that pixel intensities do not have a linear relationship between the image pair. This disparity reflects real scenarios where significant radiometric differences exist for the same objects in multisensor optical RS images.

In Fig. 4(a) and (c), we plot $\{g_q, q = 1, 2, 3, 4\}$ in (25) as a function of parameter θ under different values of λ,

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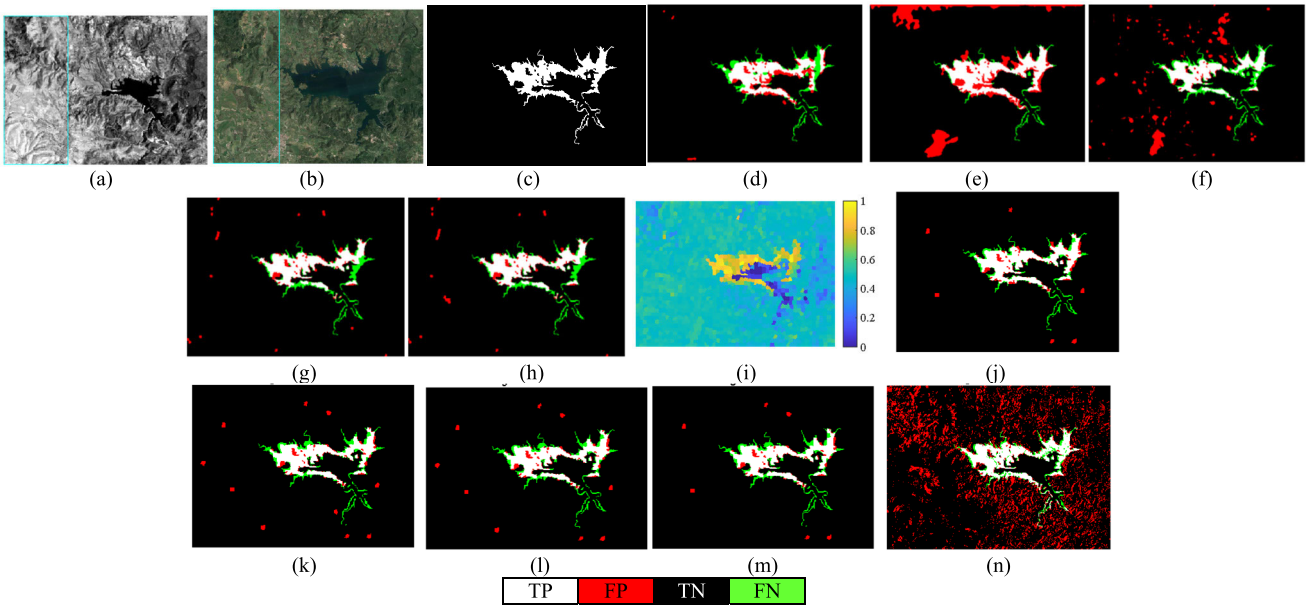


Fig. 5. Different images and BCMs of different methods on dataset 1. (a) Preevent image. (b) Postevent image. (c) Ground truth of the change mask. (d) BCM of M3CD. (e) BCM of FPMS. (f) BCM of NPSG. (g) BCM of IRG-McS.dist. (h) BCM of IRG-McS.sim. (i) Different image of SCOPS. (j) BCM of SCOPS with the selected Clayton copula. (k) BCM of SCOPS with Gaussian copula. (l) BCM of SCOPS with Student-t copula. (m) BCM of SCOPS with Gumbel copula. (n) BCM of SCOPS at the pixel level.

where $q = 1, 2, 3, 4$ represent Gaussian copula, Student-t copula, Clayton copula, and Gumbel copula. In these cases, the copula selection strategy outlined in (25) recommends the Clayton copula for CD. Accordingly, in Fig. 4(b) and (d), we depict the value of area under the curve (AUC) as a function of θ under varying values of λ when different copula functions are adopted to fit the data of the simulated multisensor optical images. Here, AUC is computed based on the receiver operating characteristic (ROC) curves obtained by thresholding the different images and a larger value of AUC implies a better performance. Since the simulated images are in accordance with Prior 1, the different images are calculated according to (14). Through Fig. 4(a)–(d), it is apparent that a larger value of g_q implies a better CD performance and that the selected Clayton copula yields an enhanced performance. In other words, a superior CD performance is attained by the copula function chosen by SCOPS compared with other candidates. Besides, the CD performance improves with an increasing value of θ, which aligns with our intuition. A higher value of θ suggests a stronger correlation between multisensor optical images, making it easier to identify changes by distinguishing between the dependence and independence structure of the features extracted from the image pair at the same location, as done in SCOPS. Fig. 4 also shows that even when training data is reduced from 40% to 20% of data in the unchanged area, the detection performance of the proposed method changes only marginally, demonstrating its robustness.

In the experimental results presented above, it has been assumed that the training data only come from unchanged areas, i.e., $\vartheta = 0$. In Fig. 4(e), we evaluate the effect of poorly selected training data from the changed area by plotting AUC as a function of ϑ, considering $\lambda = 20$, $\theta = 10$ and $\vartheta \geq 0$. Our observations show that only slight variations in performance result from mild variations of ϑ. This suggests that SCOPS is robust against changes in the value of ϑ. In other words, if data from the changed area are improperly included in the training data, the performance of SCOPS is not severely influenced. Nevertheless, reducing the value of ϑ can help avoid any possible adverse effects of improperly selected training data from the changed area. For optimal performance of the proposed method, we suggest selecting training data from the unchanged area, which can be realized by integrating geographical information, as mentioned before.

B. Experiments on Real RS Images

We further validate the effectiveness of SCOPS by testing it on two real RS datasets. Each dataset includes a pair of multisensor optical RS images, frequently used in recent literature for CD scenarios [12], [15], [44]. The ground truth of the change mask for the datasets has been widely used in the literature, including [12], [15], [44], for the purpose of CD. These datasets have been benchmarked and validated in numerous studies, making them reliable for our experiments. It is also important to note that our experiments focus only on images of moderate size. This is because the goal of this article is to develop a theoretically grounded method for CD involving multisensor optical images and not to carry out a large-scale experimental evaluation. In order to demonstrate the feasibility of the method to improve performance, like many other researchers with the same bent, we have chosen relatively small images. Once feasibility is established, the next step is for the practitioners to use methods such as ours on RS images of much larger regions. This approach aligns well with much of the existing work, as seen in [12], [15], and [44].

In dataset 1, the preevent image is of size $300 \times 412 \times 1$, which was a near-infrared image acquired by Landsat-5 in September 1995 over Sardinia, Italy, before a lake expansion. The postevent image is an optical image of size

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$300 \times 412 \times 3$, which was acquired by Google Earth (GEt2) at the same location after the lake expansion in July 1996. The spatial resolution is 30 m.

In dataset 2, the preevent image is of size $990 \times 554 \times 3$ and the postevent image is of size $990 \times 554 \times 1$. The former was acquired by Spot in 1999 over Gloucester, England, before flooding. The latter was acquired in 2000 after flooding, which is an normalized difference vegetation index (NDVI) image defined by the infrared band and near infrared band. The spatial resolution is 25 m.

When SCOPS is implemented, the number of components in the Gaussian mixture model is fixed at $Q_{\mathrm{GM}} = 3$ for the estimation of the marginal pdfs. Unless specified, during the process of K-means clustering, we fix $\alpha = 3$. Since the two images are acquired by different optical sensors and the intensities in two images may depend on different kinds of physical properties of the objects, the intensities at the same location in two images are not necessarily positively correlated. For example, with the first sensor, the backscatter intensities of clear water under windy conditions could be very high. In contrast, with another sensor, clear water has a very high absorption rate for the visible wavelengths and thus has low intensities in the image [58], [59]. In all the experiments, copulas are adopted to capture the positive dependence structure. To this end, Kendall's τ [24] is calculated based on the training patches. If it is a negative value, which indicates that there exists a negative dependence structure between the features extracted from the images, we define

$$\tilde{F}^0_{Y,c_2}(h) \approx 1 - \frac{1}{N_T}\sum_{i=1}^{N_T} \chi\left[h - \hat{h}_Y(i, c_2)\right] \quad \forall c_2. \tag{33}$$

In this case, based on (33), when SCOPS is implemented, we replace F^0_{Y,c_2} with $\tilde{F}^0_{Y,c_2}$ to guarantee that copulas are employed to characterize a positive dependence structure between $F^0_{X,c_1}[h_X(i, c_1)]$ and $\tilde{F}^0_{Y,c_2}[h_Y(i, c_2)]$.

To quantify the CD performance of different methods for an explicit comparison, we adopt three popular quantitative performance measures: Kappa coefficient (KC), F-measure (Fm), and overall accuracy (ACC), which are expressed as follows:

$$\begin{aligned} \mathrm{KC} &= \frac{2(\mathrm{TP} \times \mathrm{TN} - \mathrm{FN} \times \mathrm{FP})}{(\mathrm{TP} + \mathrm{FP}) \times (\mathrm{FP} + \mathrm{TN}) + (\mathrm{TP} + \mathrm{FN}) \times (\mathrm{FN} + \mathrm{TN})} \\ \mathrm{Fm} &= \frac{2\mathrm{TP}}{2\mathrm{TP} + \mathrm{FP} + \mathrm{FN}}, \quad \mathrm{ACC} = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + \mathrm{FN}}. \end{aligned} \tag{34}$$

In (34), TP, TN, FP, and FN represent true positive, true negative, false positive, and false negative. Larger values of KC, Fm, and ACC imply a better CD performance. To reduce the effect of randomness, under each setting, K-means clustering is implemented three times, and the averaged results of the quantitative performance measures are shown.

1) Experiments on Dataset 1: Both images are registered before further processing, as shown in Fig. 5(a) and (b). The ground truth of the change mask is given in Fig. 5(c). We randomly select a pair of patches in the unchanged area as training data, which are shown in the blue boxes in Fig. 5(a) and (b). Unless otherwise specified, the sizes of the training patches are fixed at $300 \times 130 \times 1$ and $300 \times 130 \times 3$. The number of superpixels in the training patches is $N_T = 1000$. The number of superpixels in the image pair is $N_S = 2000$.

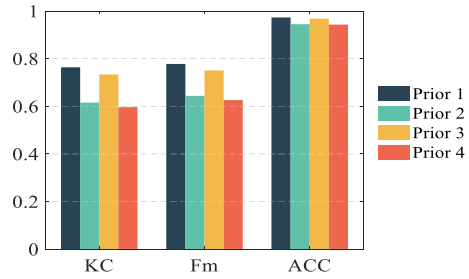


Fig. 6. Quantitative performance measures of SCOPS on dataset 1.

To test the CD performance of SCOPS under the four priors, we present KC, Fm, and ACC under different priors in Fig. 6. Based on the strategy provided in Section III-C, the optimal copula function for all channel pairs is selected to be Clayton copula function. It can be seen that a better performance is attained under Prior 1 and Prior 3. The rationality behind their superiorities can be explained as follows.

1) From Fig. 5(a), in the preevent image, both the changed area and the unchanged area contain the same type of terrain objects. In this case, the features extracted from the changed area and the unchanged area in the preevent image have similar marginal pdfs, which is consistent with Prior 1. In contrast, in the postevent image, the changed area and the unchanged area contain different types of objects. The former includes the lake, while the latter includes both the lake and objects in the land area. In this case, the marginal pdfs of the features extracted from the changed area and the unchanged area are dissimilar, which is inconsistent with Prior 1.
   In the presence of the mismatch between Prior 1 and the postevent image of dataset 1, the different image derived from Prior 1 neglects the difference between the marginal pdfs of the changed area and the unchanged area in the postevent image, and it solely uses the dependence structure in the unchanged area to detect the changes. The dependence structure in the unchanged area is captured appropriately by the selected copula function, which is distinguishable from the independence structure in the changed area. As a result, satisfactory CD performance is attained under Prior 1. The above results demonstrate that Prior 1 is quite robust to the occurrence of mismatch.
2) Prior 3 is also consistent with the fact that the features extracted from the changed area and the unchanged area in the preevent image have similar marginal pdfs. Different from Prior 1, Prior 3 takes it into consideration that the marginal pdfs of the features in the changed area and in the unchanged area are different in the postevent image. In this vein, an uninformative prior is adopted under Prior 3 to increase robustness to the variations of the marginal pdfs of the features in the changed area of the postevent image. Under Prior 3, using the marginal pdf of the features in the unchanged area in the postevent

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TABLE II
QUANTITATIVE PERFORMANCE MEASURES ON DATASET 1

Methods	KC	Fm	ACC
M3CD	0.669	0.689	0.963
FPMS	0.552	0.588	0.925
NPSG	0.559	0.587	0.947
IRG-McS.dist	0.739	0.754	0.971
IRG-McS.sim	0.733	0.749	0.971
C3D	-	-	0.925
DFR-MT	-	-	0.975
ALSC	-	-	0.965
PSGM	-	-	0.961
MDS	-	-	0.942
AFL-DSR	-	-	0.929
RMN	-	-	0.847
SCOPS with the selected Clayton copula	**0.764**	**0.778**	**0.973**
SCOPS with Gaussian copula	0.730	0.745	0.970
SCOPS with Student-t copula	0.733	0.748	0.971
SCOPS with Gumbel copula	0.743	0.757	0.972

image and the dependence structure captured by the selected copula function, the changes can be highlighted, as shown in (16).

Next, in order to verify the effectiveness of the proposed copula selection strategy, we compare the CD performance of SCOPS when different copula functions are employed. In Fig. 5(j)–(m), we provide the BCMs of SCOPS with different copula functions adopted for the first channel pair. Here, in all the BCMs, the second and the third channel pairs adopt the selected Clayton copula. It can be clearly seen that compared with the BCM of SCOPS with the selected Clayton copula, SCOPS with other copulas generates obviously more false alarms, which indicates performance degradation induced by copula misspecification. Accordingly, in Table II, we present the quantitative performance measures of SCOPS with different copulas employed for the first channel pair. We can see that SCOPS with the selected Clayton copula attains the best CD performance among all the others.

After that, we compare the performances of SCOPS under Prior 1 and the state-of-the-art methods to demonstrate the superiority of the proposed method. To this end, the BCMs of some state-of-the-art methods are presented in Fig. 5(d)–(h), which are directly quoted from [12]. Those methods include Markov model for multimodal CD method (M3CD) [22], fractal projection and Markovian segmentation-based method (FPMS) [41], nonlocal patch similarity graph-based method (NPSG) [42], iterative robust graph and Markovian cosegmentation method with distance criterion (IRG-McS.dist) [12] and iterative robust graph and Markovian cosegmentation method with similarity criterion (IRG-McS.sim) [12]. The best results of the different image and the BCM of the proposed method are shown in Fig. 5(i) and (j), respectively. By comparing the BCMs of different methods, it can be seen that the changes are identified by SCOPS more precisely. In other words, we can see that the changed and unchanged areas can be well detected by SCOPS with fewer false negatives and false positives.

Table II shows the quantitative performance measures of SCOPS under Prior 1 and other state-of-the-art methods on dataset 1. Besides the aforementioned methods, the results obtained by some other state-of-the-art methods are also presented, including C3D [20], deep feature representation and mapping transformation-based method (DFR-MT) [43], adaptive local structure consistency-based method (ALSC) [44], patch similarity graph matrix method (PSGM) [45], multidimensional scaling based method (MDS) [46], anomaly feature learning based deep sparse residual model (AFL-DSR) [47], and reliable mixed-norm based method (RMN) [48]. The results in Table II are quoted from [12] except those of C3D. The results of C3D are obtained following the steps in [20], where Ali-Mikhail-Haq copula is adopted due to its tractable closed-form expression and its performance gain demonstrated in [20], and the Kullback–Leibler divergence is calculated via the function provided by Predictive Maintenance Toolbox of MATLAB. It is worth noting that, in Table II, the baseline methods include both the supervised methods, such as C3D, and the unsupervised methods, such as IRG-McS.dist and IRG-McS.sim.

It can be seen from Table II that SCOPS attains the best CD performance among all the methods except DFR-MT. As mentioned in Section I, the interpretability and situational awareness of deep learning methods, such as DFR-MT, need to be improved, and it is very time-consuming to run deep learning methods when the computational resources are limited. From this perspective, SCOPS has obvious advantages over DFR-MT, which is that it is interpretable, reliable, and computationally efficient at the cost of negligible performance degradation. It is worth noting that although SCOPS, IRG-McS.sim, and IRG-McS.dist produce very similar ACC values, the KC and Fm values are significantly higher with SCOPS. This suggests that compared to other methods, our proposed method results in fewer false negatives and false positives. Compared with C3D, which also models the dependence structure between the image pair with copulas, the performance gain of SCOPS over C3D is due to the following facts. First, copula theory is employed by SCOPS in a different way combined with a likelihood ratio test based on the statistical model of hypothesis testing. The optimality of the likelihood ratio test contributes to precisely highlighting the changes. Second, SCOPS is superpixel-based, and it is able to maintain the structure of the objects in the changed area, which enhances the CD performance. Third, for SCOPS, the relationship between the CD performance and the copula functions is analyzed explicitly, and a strategy for determining the best copula function is provided, which guarantees a good CD performance.

To show that SCOPS only requires very limited prior knowledge, in Fig. 7, we plot KC, Fm, and ACC as a function of the width N_1 of the training patches, where the height of the training patches is fixed at $M_1 = 300$. The number of superpixels contained in the training patches varies in proportion to the width N_1 to ensure that the size of the

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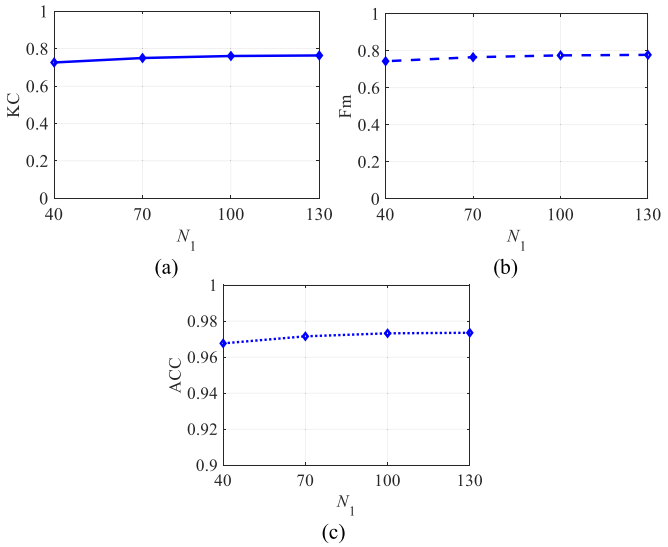


Fig. 7. Quantitative performance measures of SCOPS as a function of the width of the training patches on dataset 1. (a) KC as a function of N_1. (b) Fm as a function of N_1. (c) ACC as a function of N_1.

superpixels remains roughly invariant. It can be seen that, with less training data, i.e., with a smaller N_1, the CD performance deteriorates but at a very slow speed. We can then conclude that the performance of SCOPS is quite robust to the variations of the size of the training data and that SCOPS only requires very little training data to attain a fairly good performance.

To gauge the computational efficiency of SCOPS, we compare its processing time with the iterative robust graph and Markovian cosegmentation method (IRG-McS). Both methods were tested on a Windows desktop running MATLAB R2022b, equipped with a 12th Gen Intel[6] Core[7] i9-12900H CPU and 32 GB of RAM. The IRG-McS method took 5.738 s to run, while SCOPS finished in 4.386 s. Given that [12] demonstrated IRG-McS to be considerably more efficient than M3CD, FPMS, and NPSG, this implies that SCOPS is even more computationally efficient than these baseline methods.

Finally, to examine the impact of employing superpixels in SCOPS on CD performance compared to another case where pixels are employed as the basic processing units in SCOPS, we provide the BCM of the proposed method where CD is performed at the pixel level in Fig. 5. By comparing Fig. 5(j) and (n), it can be clearly observed that more false alarms are generated when CD is performed at the pixel level. This is because superpixels are able to maintain the structure of the objects in the images, which contributes to reducing the false alarms induced by noise and other interferences at the pixel level.

2) Experiments on Dataset 2: The registered images are shown in Fig. 8(a) and (b). The ground truth of the change mask is given in Fig. 8(c). A pair of patches is randomly selected from the unchanged area to be the training data, which are shown in the blue boxes in Fig. 8(a) and (b). Unless otherwise specified, the size of the training patches is 990×50, the number of superpixels in the training patches

[6]Registered trademark.
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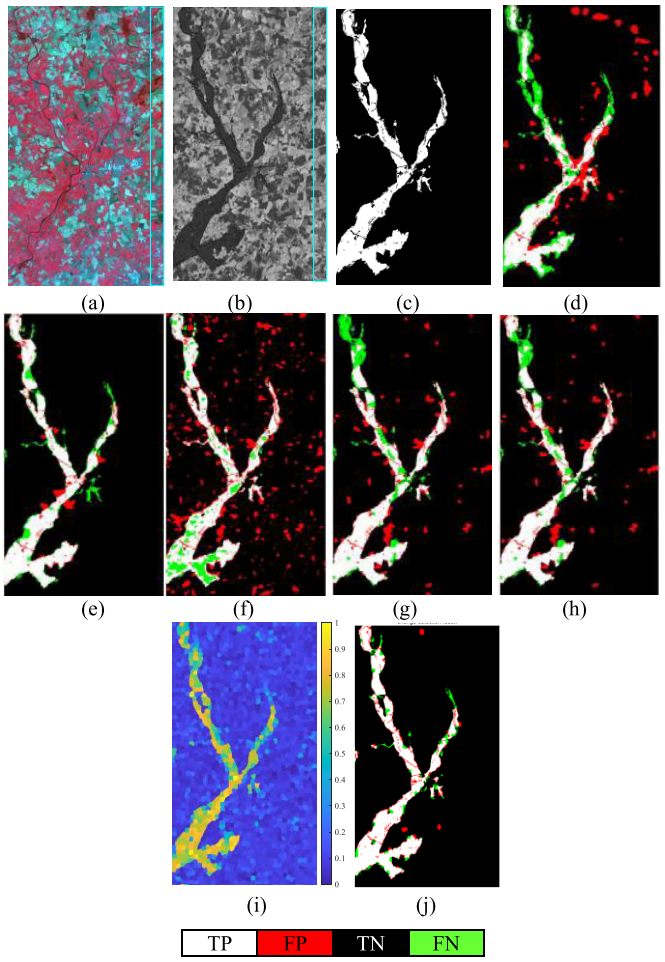


Fig. 8. Different images and BCMs of different methods on dataset 2. (a) Preevent image. (b) Postevent image. (c) Ground truth of the change mask. (d) BCM of M3CD. (e) BCM of FPMS. (f) BCM of NPSG. (g) BCM of IRG-McS.dist. (h) BCM of IRG-McS.sim. (i) Different image of SCOPS. (j) BCM of SCOPS.

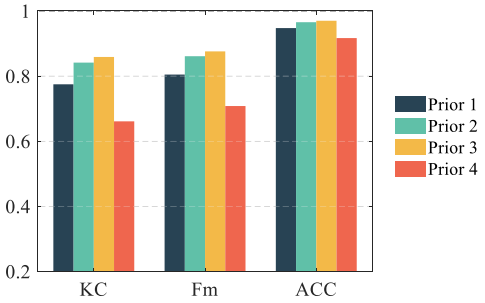


Fig. 9. Quantitative performance measures of SCOPS on dataset 2.

is $N_T = 300$ and the number of superpixels in the image pair is $N_S = 2000$.

In order to test the performance of SCOPS under different priors on dataset 2, in Fig. 9, the quantitative performance measures are presented under the four priors. Based on the strategy provided in Section III-C, the optimal copula function for all channel pairs is selected to be the Student-t copula function. Fig. 9 implies that under Prior 3, the best CD performance is achieved. This is because Prior 3 is consistent with the fact that the features extracted from the changed area and the unchanged area in the preevent image have similar marginal pdfs. Besides, it can be seen from Fig. 8(b) that the

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TABLE III
QUANTITATIVE PERFORMANCE MEASURES ON DATASET 2

Methods	KC	Fm	ACC
M3CD	0.588	0.636	0.915
FPMS	0.816	0.837	0.962
NPSG	0.608	0.663	0.902
IRG-McS.dist	0.714	0.749	0.939
IRG-McS.sim	0.735	0.768	0.942
C3D	-	-	0.899
HPT	-	-	0.957-0.964
AFL-DSR	-	-	0.836
MDER	-	-	0.818
SCOPS	**0.859**	**0.876**	**0.970**

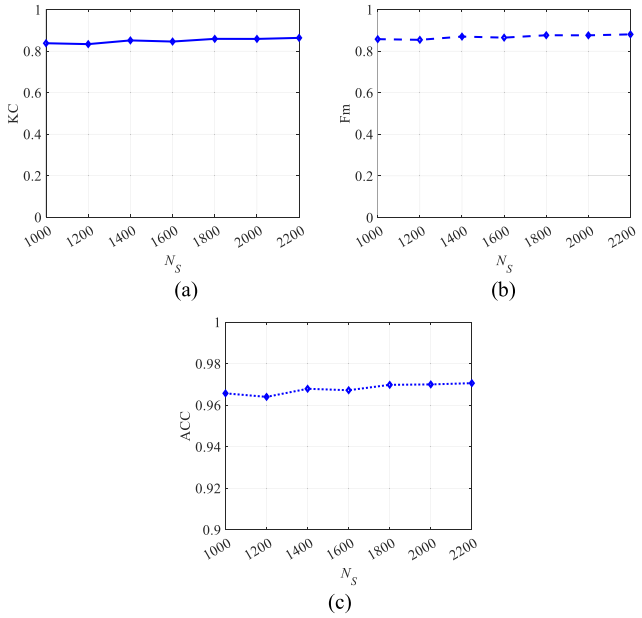


Fig. 10. Quantitative performance measures of SCOPS as a function of the number of superpixels in the image pair on dataset 2. (a) KC as a function of N_S. (b) Fm as a function of N_S. (c) ACC as a function of N_S.

changed area and the unchanged area are significantly different in the postevent image, and the marginal pdfs of the features extracted from them are very dissimilar. In this case, it is rational to apply Prior 3 and to adopt an uninformative prior for the postevent image instead of assuming the distributions of the features in the changed area and in the unchanged area in the postevent image to be the same as in Prior 1.

For comparison, the BCMs of SCOPS under Prior 3 and some state-of-the-art methods are presented in Fig. 8(d)–(h), which are directly quoted from [12]. Those methods include M3CD [22], FPMS [41], NPSG [42], IRG-McS.dist [12], and IRG-McS.sim [12]. In Fig. 8(i) and (j), we show the best results of the different image and the BCM obtained by SCOPS. Through comparison of the BCMs acquired by different methods, we can see that SCOPS is able to find the changes with higher precision, and there are fewer false positives and false negatives in the BCM generated by SCOPS compared with BCMs of the baseline methods.

Accordingly, in Table III, we present the quantitative performance measures of SCOPS under Prior 3 and the

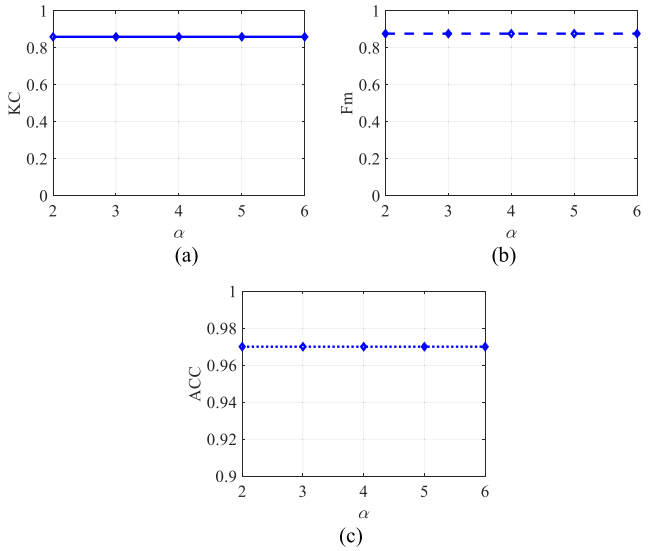


Fig. 11. Quantitative performance measures of SCOPS as a function of α on dataset 2. (a) KC as a function of α. (b) Fm as a function of α. (c) ACC as a function of α.

state-of-the-art methods, including C3D [20], M3CD [22], FPMS [41], NPSG [42], IRG-McS.dist [12], IRG-McS.sim [12], homogeneous pixel transformation method (HPT) [15], AFL-DSR [47], and multidimensional evidential reasoning based method (MDER) [49]. We directly quote the results of the state-of-the-art methods from [12] for a fair comparison. The results in Table III demonstrate that the proposed method can attain the best performance among all the methods for the CD task with multisensor images. Finally, based on the experimental results on real RS datasets in Section IV-B, in CD applications of water body monitoring, Prior 3 is quite robust, and it could be adopted by default.

To show that SCOPS is robust to varying system parameters, in Figs. 10 and 11, we plot KC, Fm, and ACC as a function of the number of superpixels, i.e., N_S, and the weight of the different image during the process of clustering, i.e., α, respectively. We can observe from Figs. 10 and 11 that the performance of SCOPS is very robust to the variations of N_S and α. It is clear that the performance of SCOPS remains stable, even when N_S and α are varied. This demonstrates that SCOPS is not overly sensitive to these parameters, reinforcing its utility and reliability for CD tasks with a high flexibility in selecting system parameters.

V. CONCLUSION

In this article, the CD problem with multisensor optical RS images was considered, and copulas were employed to model the dependence structure between the images. To overcome the drawbacks of C3D proposed in the literature, we modeled the CD problem as a binary hypothesis testing problem and proposed a new method named SCOPS, where we provided an explicit strategy for copula selection for CD tasks analytically. Finally, the effectiveness of the copula selection strategy adopted by the proposed method and the superiority of SCOPS over the state-of-the-art methods, including C3D were validated on simulated data and real RS datasets. In the future, we intend to extend the application of SCOPS to fully unsupervised scenarios. It is also worth pointing out

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that our work in this article is more of a feasibility study, which explores the possibility of capturing the dependence structure between multisensor optical RS data by selected copulas. It can be seen that the features used in this article are very simple, which is the mean of the pixel intensities in a superpixel. In the future, we would also attempt to extract more complex features from the RS images to further improve the CD performance. Another limitation of this work is that we have only verified that the proposed method produces advantages on multisensor optical image pairs, i.e., RS images obtained by different optical sensors. In the future, we would also run more experiments on RS datasets, which contain multisource image pairs, i.e., images acquired by different types of sensors, including, for example, optical sensors and synthetic-aperture radars.

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